SAMUEL J. TUSICK

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SUMMARY

Software Engineering student who builds AI and automation systems that teams use daily: an institute-wide automation stack, an LLM-driven virtual patient platform with chat, voice, and avatar modes on serverless AWS, and a machine learning payload for a high-power rocket. U.S. citizen, eligible to pursue a security clearance.

EDUCATION

Florida Gulf Coast University, B.S. Software Engineering *Expected May 2027*

- In-Major GPA 3.83 | Dean's List (2023, 2024, 2025) | Campbell Engineering and Kunkle Family Endowed Scholarships
- Coursework: Machine Learning, AI Agent Studio, Software Architecture, Data Structures & Algorithms, Data Engineering, Cloud Computing, Software Engineering Fundamentals, Requirements Engineering & Analysis

TECHNICAL SKILLS

- **Languages:** Python, C++, C, SQL, JavaScript
- **AI and Automation:** LLM API integration, prompt and skill design, agentic workflow automation, orchestration of multi-service AI systems, Neural Networks and genetic algorithms
- **Cloud and Backend:** AWS (API Gateway, Lambda, Aurora, Cognito, IAM, CDK), Flask, REST APIs, PostgreSQL, Supabase, SQLite
- **Frontend and Dev Tools:** React, Vite, Tailwind CSS, Git, GitHub Actions, Selenium, Jira, Azure DevOps, OpenRocket

EXPERIENCE

FGCU Dendritic AI & Data Science Institute, Student Research Assistant *Mar 2025 to Present*

- Built an LLM-powered pipeline for AI avatar course production: one skill turns professor PDFs and slides into scripts, and a second generates the avatar video and matching slide deck from each script and institute standards. 45% efficiency gain, used by the dev and infrastructure teams.
- Automated operations with scheduled and event-driven workflows: a 9 AM Jira daily briefing posted to Microsoft Teams for the whole institute, and file-upload alerts for the media and infrastructure teams. Also built a reusable skill that generates the institute handbook of team workflows, used institute wide.
- Assisted coordination and execution of the 2026 FGCU Summer AI Academy (500+ participants, 150+ instructional hours) and built 14 curriculum-specific AI tutors for the Engineering Learning Hub.

PROJECTS

MAIVA: AI Virtual Mental Health Patient Training Platform | AWS, React, Vite, LLM API

- Co-own architecture, data modeling, and technology decisions, turning requirements from two clinical assistant professors into a platform where students practice clinical interviews with AI-simulated patients by chat, voice, or 3D avatar. Chat is in active testing with real mental health professionals; voice is in development.
- Designed the orchestrator that connects the UI, speech-to-text, text-to-speech, and LLM API for every mode. It streams the patient's reply sentence by sentence over a WebSocket, saves each turn to the database in real time, and rebuilds a session from the saved transcript after a restart.
- Built speech-to-text capture and the safety-escalation flow that pauses the session clock so the student can resume or cancel. Refactored the React frontend and run Local, Dev, and Test environments on a serverless AWS backend.

Real-Time Landing Site Detection with ML | Python, Raspberry Pi, OpenRocket

- Team lead and software and hardware developer for a 4-person team selected as one of two winners of the FGCU Payload Challenge, earning funding and a flight on the club's high-power rocket.
- Designing a camera-only payload: a segmentation model labels terrain safe or unsafe live on a Raspberry Pi with IMU, barometer, and GPS logging, tested during powered ascent to stress it with vibration and motion blur.
- Built an OpenRocket flight simulation and a pretrained-model proof of concept. Targets: 2+ predictions per second, 80% pixel accuracy, reported either way.

ML Self-Driving Car Simulator | Python, Pygame, Gymnasium

- Building a neural network and genetic algorithm trainer from scratch with no ML libraries. Built the simulation: car physics, pixel-mask collision, five ray-cast sensors, checkpoints, and a Gymnasium environment.
- Baseline before training: a random policy finished 0 of 1,000 episodes, and only 2 passed even one checkpoint.

Switch4Good Database System | PostgreSQL, SQL, Data Engineering

- Enabled data-driven nonprofit operations at scale, supporting 100+ universities and 1,500+ students, by leading development of a PostgreSQL database and schema for participation, projects, partnerships, and outcomes.

LEADERSHIP AND CERTIFICATION

Deeptech Club, Payload Team Lead | FGCU AI Day, Panelist *June 2026 to Present*

Team Lead, Switch4Good Database and Pinterest Testing Suites | Subteam Lead, Eagle CyberNest Lab *2025 to 2026*

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate *2025*